

Corporate Finance

Part 2. Estimation and Application of the structural corporate finance model

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Abstract

This project investigates the empirical estimation of a structural dynamic corporate finance model characterized by capital adjustment costs and stochastic productivity. The model is estimated using the Generalized Method of Moments (GMM) and the Simulated Method of Moments (SMM), implemented in TensorFlow. GMM exploits orthogonality conditions derived from the Euler equation, while SMM matches simulated model moments to empirical moments. Estimation is conducted using synthetically generated data to evaluate parameter recovery, moment matching, and statistical validity. In addition, a Bayesian estimation using Hamiltonian Monte Carlo (HMC) is implemented for comparison.

1 Introduction

Dynamic corporate finance models provide a structured framework to analyze firm behavior under uncertainty. These models derive investment decisions from explicit optimization problems and impose internally consistent economic restrictions. However, theoretical consistency alone is not sufficient. For the model to be empirically meaningful, its structural parameters must be estimated using data. Structural estimation allows recovery of deep parameters such as adjustment costs, shock persistence, and productivity volatility. These parameters govern the dynamic responses of firms to economic shocks and determine the quantitative relevance of the model's mechanism [16].

Estimation in dynamic corporate finance models is challenging for several reasons. First, the model is nonlinear and forward-looking. The value function is not available in closed form. Second, optimal investment decisions depend on expectations about future productivity. These expectations must be approximated numerically. Third, structural parameters enter both the policy function and the equilibrium conditions in a non-trivial manner. As discussed by Strebulaev and Whited [16], empirical implementation therefore requires specialized econometric techniques that respect the model's structure.

Two widely used approaches in this literature are the Generalized Method of Moments (GMM) and the Simulated Method of Moments (SMM). GMM, formalized by Hansen [8], exploits orthogonality conditions implied by the model's first-order optimality conditions. In dynamic settings, these typically arise from Euler equations. Under valid instruments, GMM

provides consistent parameter estimates and allows formal specification testing through over-identifying restrictions. SMM, in contrast, matches moments generated by simulated model data to empirical moments. Its asymptotic properties are studied by Pakes and Pollard [14]. SMM is particularly useful when the model is complex and analytical moment conditions are difficult to derive. This report implements both GMM and SMM for the basic dynamic investment model. Estimation is conducted using synthetically generated data to evaluate parameter recovery in a controlled environment.

2 Literature Review

Structural estimation in dynamic economics began with the development of moment-based methods that linked optimization theory with empirical implementation. Hansen [7] formalized the Generalized Method of Moments, providing a systematic framework to estimate parameters using orthogonality conditions implied by rational expectations. Hansen and Singleton [8] demonstrated how intertemporal Euler equations could be estimated directly without specifying full likelihood functions. These developments influenced macroeconomics and asset pricing, where dynamic optimization models became empirically tractable. The theoretical foundations of simulation-based estimators were later established by Pakes and Pollard [14] and extended by Duffie and Singleton [5] and Gourieroux et al. [6]. These contributions enabled the estimation of nonlinear dynamic models through simulated objective functions.

Structural estimation entered corporate finance through dynamic investment models with adjustment costs. Early applications relied on the Euler equation-based GMM to estimate parameters governing capital accumulation. Whited [19] provided one of the first such applications in corporate finance, followed by Bond and Meghir [1] and Love [11]. These studies used moment conditions derived from investment optimality to measure financing frictions and adjustment costs. However, concerns regarding measurement error and local approximation led to methodological refinement. As computational methods improved, simulation-based approaches such as the Simulated Method of Moments (SMM) gained prominence. Hennessy and Whited [18] employed SMM to estimate dynamic corporate models with richer nonlinearities, demonstrating the advantages of matching global model moments rather than relying solely on local Euler conditions.

More recent developments have expanded structural estimation beyond moment-based techniques. Likelihood-based approaches, inspired by Rust [15], and Bayesian methods using Markov Chain Monte Carlo have been applied to dynamic corporate settings. Studies such as Korteweg [10] and DeAngelo et al. [4] illustrate how structural estimation can quantify economic primitives and enable counterfactual analysis. Strebulaev and Whited [16] provide a comprehensive synthesis of these methodological advances within corporate finance. The evolution of the literature thus reflects a progression from Euler-based GMM to simulation-based and likelihood-based frameworks, with increasing emphasis on computational robustness and identification in dynamic investment models.

3 Methodology

We consider the discrete-time dynamic investment model by Strebulaev and Whited [16]. The representative firm chooses next-period capital to maximize the present value of shareholder payouts. The capital accumulation equation is given by

$$k_{t+1} = (1 - \delta)k_t + I_t, \quad (1)$$

where δ denotes the depreciation rate and I_t is investment.

Output is produced according to

$$\pi(k_t, z_t) = z_t k_t^\theta, \quad (2)$$

where z_t follows an AR(1) process:

$$\log z_{t+1} = \rho \log z_t + \varepsilon_{t+1}, \quad \varepsilon_{t+1} \sim N(0, \sigma^2). \quad (3)$$

Investment adjustment costs are quadratic:

$$\psi(I_t, k_t) = \frac{\psi_0}{2} \left(\frac{I_t}{k_t} \right)^2 k_t. \quad (4)$$

The firm's optimal investment decision implies an Euler equation of the form

$$1 + \psi_I(I_t, k_t) = \mathbb{E}_t \left[\frac{\pi_k(k_{t+1}, z_{t+1}) - \psi_k(I_{t+1}, k_{t+1}) + (1 - \delta)(1 + \psi_I(I_{t+1}, k_{t+1}))}{1 + r} \right]. \quad (5)$$

Structural estimation aims to recover the parameter vector

$$\Theta = (\theta, \rho, \sigma). \quad (6)$$

The depreciation rate δ , adjustment cost parameter ψ_0 , and interest rate r are calibrated and held fixed. Estimation is performed using synthetically generated data with a burn-in period to eliminate dependence on initial conditions. Two estimation approaches are implemented: GMM and SMM.

3.1 Generalized Method of Moments (GMM)

GMM estimates parameters by exploiting orthogonality conditions implied by the model's first-order optimality conditions. In this context, the Euler equation generates moment restrictions of the form

$$\mathbb{E}[u_t Z_t] = 0, \quad (7)$$

where u_t is the Euler residual and Z_t is a vector of instruments belonging to the firm's information set.

Let

$$u_t(\Theta) = (1 + \psi_I(I_t, k_t)) - \frac{\pi_k(k_{t+1}, z_{t+1}) - \psi_k(I_{t+1}, k_{t+1}) + (1 - \delta)(1 + \psi_I(I_{t+1}, k_{t+1}))}{1 + r}. \quad (8)$$

The sample moment vector is constructed as

$$g_T(\Theta) = \frac{1}{T} \sum_{t=1}^T \begin{bmatrix} u_t z_t \\ u_t k_t \\ u_t \\ \text{Corr}(\log z_t, \log z_{t-1}) \\ \text{Var}(\log z_t) \end{bmatrix}. \quad (9)$$

The last two moments incorporate the productivity process directly.

The GMM objective is

$$Q_T(\Theta) = g_T(\Theta)' W g_T(\Theta), \quad (10)$$

where W is the weighting matrix.

3.2 Simulated Method of Moments (SMM)

SMM estimates parameters by matching model-implied moments to empirical moments. Unlike GMM, it does not rely directly on Euler equation residuals. Instead, it focuses on reproducing observable investment and productivity dynamics.

Let m^{data} denote moments computed from synthetic data and $m^{sim}(\Theta)$ denote moments from simulated data under candidate parameters.

The moment vector includes:

$$m(\Theta) = \begin{bmatrix} \mathbb{E}[I_t/k_t] \\ \text{Corr}(I_t/k_t, z_t) \\ \text{Corr}(\log z_t, \log z_{t-1}) \\ \text{Var}(\log z_t) \end{bmatrix}. \quad (11)$$

The SMM objective is

$$Q_T(\Theta) = (m^{data} - m^{sim}(\Theta))'(m^{data} - m^{sim}(\Theta)). \quad (12)$$

4 Answers to the Questions

Q1. Section 4 of the paper introduced two types of estimation methods, GMM and SMM. Please implement the two types of estimation methods in Tensorflow and compare the effectiveness of two methods. Again, you can start by discussing how effectiveness is defined and what measures can be used to evaluate effectiveness.

Answer 1:

The effectiveness of the GMM and SMM estimators is evaluated along four main dimensions:

- (a) **Parameter Recovery:** The estimated parameters should be close to the true parameter values. The true and estimated parameter values for GMM and SMM are reported in Table 1.
- (b) **Moment Matching:** The model-implied moments should closely match the empirical moments. The comparison of the data and model-implied moments for GMM and SMM estimation is provided in Tables 2 and 3, respectively.
- (c) **Objective Function Convergence:** The loss function should decline smoothly and stabilize at a low value. Figure 1 presents the convergence curves of the objective function for GMM and SMM structural estimation.
- (d) **Statistical Validity (J-statistic and p-value):** Over-identification tests should not reject the model restrictions. Under the null hypothesis that the model is correctly specified, the J-statistic follows a chi-square distribution with degrees of freedom equal to (number of moments - number of parameters). If the p-value is greater than 0.05, the model restrictions cannot be rejected. If the p-value is less than 0.05, the model is rejected. Table 4 reports the loss function values and the J-statistic and corresponding p-values for the GMM and SMM estimation.

In percentage terms, the deviations in parameter estimation are very small for both methods. In structural estimation, small absolute deviations indicate strong identification. The SMM estimator recovers the persistence and volatility parameters almost exactly. GMM shows a slightly larger deviation in ρ , but the error remains modest. An objective function value close to zero implies that the orthogonality conditions are well satisfied. The reported GMM loss of 0.000053 is very small, indicating tight moment alignment. For SMM, the reported value is effectively very close to zero. This shows that the simulated model moments match the data moments almost perfectly. The J-statistic further evaluates model validity. Under correct specification, a high p-value implies that the imposed restrictions cannot be rejected. The GMM p-value of 0.9659 and the SMM p-value of 0.9872 both indicate strong statistical validity. Values above 0.05 are generally acceptable.

For GMM, Euler-based moments and productivity moments show only minor differences between data and model values. This confirms that the first-order optimality conditions are approximately satisfied in the simulated data. For SMM, the selected investment and productivity moments match almost exactly. The loss curves also show stable convergence for both methods. GMM converges rapidly and stabilizes early. SMM declines steadily and reaches a lower final loss. Overall, both estimators are effective in this controlled experiment. GMM is computationally efficient and performs well under Euler-based restrictions. SMM achieves slightly stronger global moment matching, though at a higher computational cost.

Q2. What are the potential applications of the risky debt corporate finance model in section 3.6 of Strebulaev(12)?

Table 1: True parameter values and estimated parameters obtained from GMM and SMM.

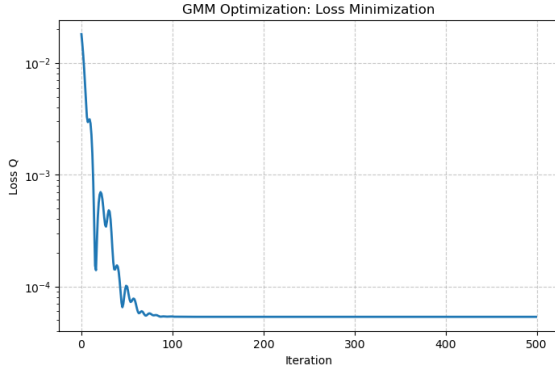
Parameter	True	GMM Estimated	SMM Estimated
θ	0.6000	0.6018	0.6042
ρ	0.9000	0.9105	0.8995
σ	0.1000	0.0976	0.1000

Table 2: Comparison of data and model-implied moments used in the GMM estimation.

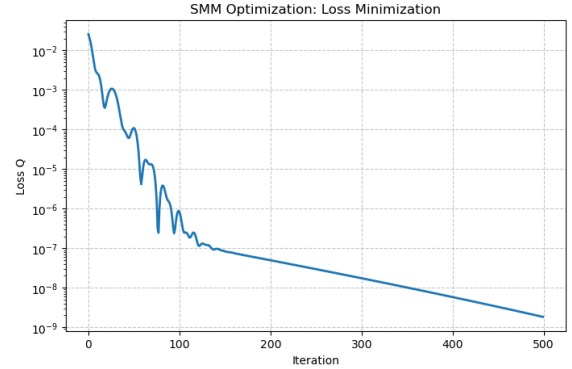
Moment	Data	Model
Mean Euler Residual	-0.8566	-0.8626
Euler $\times$ Productivity	-0.2755	-0.2766
Euler $\times$ Capital	-0.8072	-0.8085
Corr($\log z_t, \log z_{t-1}$)	0.9070	0.9059
Var($\log z$)	0.0551	0.0514

Table 3: Comparison of empirical and simulated moments used in the SMM estimation.

Moment	Data	Model
Mean(I/k)	0.0991	0.0982
Corr($I/k, z$)	0.6918	0.6925
Corr($\log z_t, \log z_{t-1}$)	0.9048	0.9044
Var($\log z$)	0.0534	0.0536



(a) GMM optimization loss curve



(b) SMM optimization loss curve

Figure 1: Training loss diagnostics for the GMM and SMM estimation.

Table 4: Loss function values and over-identification test statistics for GMM and SMM estimators.

Statistics	GMM	SMM
Loss Q	0.000053	0.000001
J-statistic	0.069400	0.001012
p-value	0.965900	0.987200

Answer 2:

- (a) **Quantifying the cost of external financing:** This application uses a dynamic model with risky debt and the possibility of default to measure how costly it is for firms to raise outside funds. Instead of assuming these costs, the model estimates them from the data by matching the firm's observed behavior, such as investment, borrowing, and default decisions [9].
- (b) **Investment–cash flow sensitivities:** This application uses the risky debt model to study whether investment–cash flow sensitivity is really a good measure of financial constraints. In the model, firms choose how much to invest and borrow, knowing that they may default if income becomes too low. Default occurs when equity holders prefer to walk away rather than invest more in the firm [13].
- (c) **Studying agency conflicts and leverage dynamics:** This application looks at how conflicts between debt holders and equity holders affect a firm's capital structure over time. The model allows firms to choose investment and financing dynamically, and default occurs when the value of equity falls to zero [17].

Q3. For the applications you mentioned in (2), what are the defects of the model? Can you provide a literature survey on potential solutions to the defects identified?

Answer 3:

The possible defects for the applications mentioned in part (2) are

- (a) **Investment Variance (Overshooting):** This is considered a defect because the basic model often generates an investment series that is significantly more volatile than what is observed in empirical data. The model assumes firms can adjust their capital stock with minimal friction, causing them to overreact to every productivity shock [9].
- (b) **Curse of Dimensionality:** This computational defect limits the realism of structural models because each new state variable (cash holdings or debt maturity) exponentially increases the complexity of the numerical solution. Because traditional algorithms struggle to track multiple variables simultaneously, researchers are often forced to omit critical firm features to keep the model solvable [17].
- (c) **Static Debt Maturity:** Many dynamic models assume a fixed maturity structure for a firm's debt, which is a defect because it ignores a firm's ability to endogenously choose when to retire or roll over its obligations. In reality, firms manage their debt maturity strategically to handle liquidity risk [17].

Some of the potential solutions identified for these defects are as follows:

- (a) **Investment Irreversibility:** This solution addresses the Investment Variance defect by incorporating irreversibility into the firm's investment cost function. By assuming that capital cannot be sold for its purchase price, the model forces firms to be more cautious and smoothens their investment behavior [2].

- (b) **Deep Learning:** This method solves the Curse of Dimensionality by using neural networks to approximate the firm's value and policy functions. Unlike traditional grid-based methods, it uses an all-in-one Bellman residual approach that can efficiently handle high-dimensional problems [12].
- (c) **Endogenous Debt Retirement:** This solution addresses the Static Debt Maturity defect by allowing firms to endogenously choose the rate at which they retire their debt. By making maturity a choice variable, the model can capture how firms strategically adjust their debt duration to manage liquidity [3].

5 Bonus Questions

5.1 Bonus Question 1

Instead of GMM and SMM, please develop a Bayesian estimation using Hamiltonian Monte Carlo or other MCMC methods (for sampling from the posterior distribution) together with filtering (for computing the likelihood).

Q1. What are your prior assumptions on the parameters?

Answer 1: The prior assumptions on the parameters are as follows:

- (a) Curvature $\theta \sim \text{Beta}(2, 2)$ where its range is restricted to $[0.2, 0.9]$. This restriction reflects standard economic theory. If θ is very close to zero, capital would have almost no effect on production. If it is very close to one, the model would imply nearly constant returns to scale, which may lead to unrealistic dynamics. Within this interval, we assume a scaled Beta(2, 2) distribution. The Beta(2, 2) distribution is symmetric and places slightly more weight on values in the middle of the range than at the extremes.
- (b) Persistence $\rho \sim \text{Beta}(2, 2)$ where its range is restricted to $[0.7, 0.99]$. Empirical evidence in macroeconomics shows that productivity shocks are usually highly persistent. Values below 0.7 would imply that shocks disappear too quickly, while values above 1 would violate the stationarity condition of the $AR(1)$ process. Within this range, we again use a scaled Beta(2, 2) distribution.
- (c) Volatility $\sigma \sim \text{HalfNormal}(0.1)$ because $\sigma > 0$. The Half-Normal distribution concentrates mass near zero while allowing larger values with decreasing probability.

Q2. What filtering method have you chosen? What are the pros and cons of your algorithm vs other algorithms?

Answer 2: For this problem, the Particle Filter (also known as Sequential Monte Carlo or SMC) is the chosen filtering method. This method is appropriate because the structural model is nonlinear and does not admit a closed-form likelihood.

Pros

- **Handles nonlinearity:** Our investment model is nonlinear due to adjustment costs and exponentiation of productivity. This method handles nonlinear dynamics naturally.
- **Non-Gaussian distributions:** It does not assume that the hidden states or errors are normally distributed.
- **Unbiased likelihood estimator:** The particle filter produces an unbiased estimator of the likelihood.

Cons

- **High computational cost:** It is significantly slower than other filters because it must run the model simulation many times for every single time step in the data.
- **Particle degeneracy:** Over time, many particles receive near-zero weight, and only a few dominate. This reduces efficiency, leading to an inaccurate estimate of the likelihood.
- **Sampling Noise:** If the number of particles is small, the likelihood estimate becomes noisy.

Q3. Please implement your Bayesian estimation method using TensorFlow Probability (TFP).

Answer 3: We implement the Hamiltonian Monte Carlo (HMC) Bayesian estimation procedure using TFP in TensorFlow.

Q4. What tests can you design to assess the validity of your estimation method?

Answer 4: The validity and performance of the Bayesian HMC estimation method are compared with the GMM and SMM estimators. Each method is repeated five times with independent random seeds to account for sampling variability. For HMC, the posterior mean of the parameter draws is used as the point estimate to ensure comparability with the point estimates produced by GMM and SMM. The estimators are compared using the following standard metrics:

- (a) **Bias:** Bias measures the average deviation of the estimator from the true parameter value. It captures systematic estimation error.
- (b) **Variance:** Variance measures the dispersion of the estimator across independent runs. It captures the stability of the estimator across different samples.
- (c) **Root Mean Squared Error (RMSE):** RMSE is the primary measure used to compare overall estimator performance.
- (d) **Computational Efficiency:** In addition to statistical accuracy, we evaluate computational performance. We compare the average time per optimizer iteration for these three methods.

Q5. How does this Bayesian estimation method compare to GMM and SMM?

Answer 5:

In terms of runtime (Table 5), GMM and SMM are computationally fast and comparable, whereas HMC is substantially slower, reflecting the high cost of repeatedly evaluating a particle-filter-based likelihood inside an MCMC algorithm. Table 6 compares the performance of the Bayesian HMC estimator against the classical GMM and SMM approaches. Overall, both GMM and SMM outperform HMC in terms of RMSE for most parameters in this implementation. For θ , HMC achieves a relatively small bias, but its RMSE is notably larger than that of GMM and SMM. Similarly, for the volatility parameter σ , HMC exhibits higher bias and RMSE compared to both GMM and SMM. For ρ , HMC performs comparably to the moment-based methods in terms of RMSE, but it does not clearly dominate them.

Table 5: Computational runtime comparison (in seconds) for one estimation run under each method (GMM, SMM, and HMC).

Model	Time(s)
GMM	5.785250
SMM	6.684671
HMC	60.302805

Table 6: Performance comparison of GMM, SMM, and HMC across five independent runs.

	θ			ρ			σ		
Metric	GMM	SMM	HMC	GMM	SMM	HMC	GMM	SMM	HMC
Bias	0.00648	0.00512	0.00394	0.00128	0.00056	0.00270	0.00498	0.00016	0.01416
RMSE	0.01104	0.01305	0.02135	0.00861	0.00621	0.00632	0.00878	0.00643	0.01569
Variance	0.00008	0.00014	0.00044	0.00007	0.00004	0.00003	0.00005	0.00004	0.00005

These results suggest that, in this finite-sample setting, the moment-based estimators (GMM and SMM) provide more accurate point estimates than HMC. However, this difference should be interpreted cautiously. Due to computational constraints, the HMC algorithm was run with a relatively small number of iterations. As a result, the posterior may not have been fully explored. In particular, insufficient MCMC iterations and the additional Monte Carlo noise introduced by the particle filter can increase variance and RMSE. Therefore, the comparatively weaker performance of HMC in some parameters likely reflects computational limitations rather than a fundamental inferiority of the Bayesian approach.

5.2 Bonus Question 2

In Section 5.3.2 of Cronqvist(24), the CEO of the firm chooses the effort level, manipulation and risk level. The board chooses the compensation structure of the CEO.

Q1. What are the key problems that the authors try to illustrate with the model?

Answer 1: The key problems that the authors try to illustrate are as follows:

- (a) The first problem they are trying to solve is whether the price drop happened only because investors were previously underestimating CEO pay and then corrected their mistake, or whether the drop was also caused by real changes inside the firm.
- (b) The authors also examine how increased investor attention can change company behavior. If investors start paying more attention to CEO pay, the board may redesign the CEO's compensation package. Changes in compensation can affect the CEO's incentives, potentially influencing effort, risk-taking, and short-term decision-making. These changes can impact future company performance.
- (c) The third problem is measuring how important this limited attention effect really was. The model allows the authors to estimate how much of the CEO pay investors were ignoring before the rule and how much of the value drop can be explained by that.

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